# An agent based model of Brenner´s theory of transition

**Abstract**

The results of the Brenner Debate produced an enhanced theory of transition from feudalism to capitalism based in rural $17^{th}$ Century England. A key tenant of this school of thought is the role of “improvement" as both a motor in the divorce between peasants and their means of subsistence, and a consequence of newly created market imperatives both landlords and tenants were now subject too. To test the role of improvement ideology in the transition an agent based model is constructed. This incorporates tenants, landlords, ecological deterioration and rental markets. The critical role of rental market pressure is modelled through a transition from customary rent to leasehold rents.

## 1 Introduction

According to [Comninel, 2000, p.10] feudalism may be defined as a mode of production in which the ruling class extracts surplus value from direct producers via non-economic means, often political and under the threat of force. In contrast, capitalism extracts surplus value from labourers, who are not direct producers and do not posses the means of production, via economic means. Capitalists must re-invest the extracted surplus value, in order to reproduce their class position, to keep up with increasing productivity. Through an analysis of the transition from one class structure to another, one might hope to gain insights into common identifiers which are associated with the collapse of a mode of production. Improvement ideology emerged in the mid $16^{th}$ Century composed of both practical methods to increase the output of a given piece of land, and as a method through which to better the condition of the English peoples, as argued by Captain Walter Blith.

### 1.1 Marx on the transition and primitive accumulation

Marx argued against the inevitability of capitalism, comparing it to the “original sin” in theology [Marx, 1894, p.507]. Where capitalism’s “origin is supposed to be explained when it is told as an anecdote of the past" [Marx, 1894, p.507]. Marx instead produced a theory based off the end result and its defining characteristics of wage co-modification and the separation of the means of production from wage-employees. For Marx the transformation of goods into capital requires a social transformation in the relations between the producer and the ruling class. This process Marx labels Primitive Accumulation. He identified the "expropriation of the self-supporting peasants" as a factor but also ”the destruction of rural domestic industry" [Marx, 1894, p.531]. The former of which is the first step in the formation of capital, drawing a distinction between feudal wealth and capital. Marx states ”what enables money-wealth to become capital is the encounter, on one side, with free workers; and on the other side, with the necessaries and materials etc, which previously were in one way or another the property of the masses who have now become object-less, and are also free and purchasable" [Marx, 2005]. Marx identifies this process of dispossession as having two components, firstly the “forcible driving of the peasantry from [their] land" and secondly a “usurpation of the common lands" [Marx, 1894, p.514]. With both factors Marx uses Francis Bacon’s “History of the Reign of King Henry VII" to pinpoint historical legislation off the back of which much of this dispossession was carried out. For Marx these tendencies contributed to a final state of a “degraded and almost servile condition of the mass of the people, the transformation of them into mercenaries, and of their means of labour into capital" [Marx, 1894, p.511]. This degradation of the peasantry can be seen in the collapse of the independent yeoman who was replaced by tenants who depended on yearly leases. Producing “a servile rabble dependent on the pleasure of the landlords" [Marx, 1894, p.511]. This available labour was now “transformed into material elements of variable capital" [Marx, 1894, p.530] thereby introducing the final components for the transition, “ the peasant, expropriated and cast adrift, must buy their value in the form of wages, from his new master, the industrial capitalist" [Marx, 1894, p.530]. For Marx it is a combination of social relations and material conditions that produces the banality of capitalist economic logic resulting in “the subjugation of the labourer to the capitalist" [Marx, 1894, p.523]. In his analysis Marx had a very limited mention of improvement language. Suggesting the ”revolution in the conditions of landed property was accompanied by improved methods of culture, greater co-operation, concentration of the means of production" [Marx, 1894, p.530]. This lack of involvement of improvement within Marx's theory of transition could in part be explained by a lack of access to many of the primary sources used by the likes of Brenner and Woods in constructing their own theories. But also speaks to the fact that Marx was first and foremost a philosopher not a historian.

### 1.2 Brenner's theory of transition

With the publication of "Agrarian Class Structure and Economic Development in Pre-Industrial Europe" in 1976 Robert Brenner produced a crucial advance in the causes of the very first case study of the transition from feudalism to capitalism. It is in this very word "transition" that Brenner argued even Marxist historians such as Paul Sweezy or Maurice Dobb had been led astray [Brenner, 1977, p.41-53], emphasising that they were assuming an a priori existence of an embryonic capitalism within feudal society.Several theories of transition may be grouped as commercial models. The keystone of these theories is that capitalism arose as the barriers for its development were removed [Wood, 2002, p.12]. Crucially, the driving factor behind this development was man's "propensity to truck, barter, and exchange" [Smith and Stewart, 1963]. This force led to the division of labour and ultimately industrial capitalism. Brenner and Wood's fundamental critique of this approach is that it assumes a nascent agricultural or industrial capitalism within other modes of production, such that there is no real transition and capitalism was inevitable [Wood, 2002, p.51]. Brenner's theory sought internal conflicts within feudalism, as opposed to an external impetus, which might produce its collapse. Wood offers an analysis that " it [was] a matter of lords and peasants, in certain specific conditions peculiar to England, involuntarily setting in train a capitalist dynamic while acting in class conflict with each other to reproduce themselves as they were" [Wood, 2002, p.52]. This focus on the class conflict nature of the transition is what distinguished Brenner and Wood's theory from other contemporary Political Marxists. Brenner identifies the internal logic of feudalism that inhibits surplus reinvestment. For the feudal lord it was often easier to find new avenues to "squeeze" [Brenner, 1976, p.74] out more absolute surplus value from their subjects instead of increasing their labour efficiency. Due to the fact that a lord's power was directly correlated to size of the population under his control this lead to a focus on labour immobility, and maximising the ease with which he could extract surplus value from his peasants by non-economic means. Resulting in restrictions on land mobility and "unfree peasants [unable] to convey their land to other peasants without the lord's permission" [Brenner, 1976, p.50]. The centuries previous to the $17^{th}$ Century were characterised by the end of serfdom in England, resulting in peasants whose lords extracted surplus value partly through the rent of their land. These were customary rents that decreased in value for the lord over time due to inflation. This loss in income revenue was counteracted through the imposition of arbitrary fines on peasants wherever land was sold or inherited. However over the $15^{th}$ and $16^{th}$ centuries landlords successfully crushed peasant land rights in order to combat this decreasing power. This decay in freehold rights meant that lord could slowly accumulate large demesne holdings, which could be leased out to larger tenants [Hoyle, 1990, p.7]. Who in turn sub-let to smaller farmers or employed wage labourers. This new wave of larger tenancies brought two new sources of market imperatives, firstly the lord had to keep his rents competitive as he was now trying to undercut the rest of the domestic tenancy market, given that the larger tenant had the ability to move away. Secondly the tenant had to compete on a goods market, requiring them to sell the products of their husbandry in order to make enough profit to fulfil market dependant rent. This required a shift from production for subsistence to production for sale. These two forces resulted in a new weakly symbiotic relation between lord and capitalist tenant in which both were driven to re-invest surplus value back into their land, thereby improving agricultural yields and transforming feudal wealth into capital. This established a new set of laws of motion in which the lord and tenant were compelled to improve their agricultural yields in order to self-replicate their class condition. Brenner claims that "agricultural improvement not only made it possible for an ever greater proportion of the population to leave the land to enter industry" [Brenner, 1976, p.67] but also mean that a larger proportion of the domestic population was no longer involved in agriculture, freeing peasants to become the wage labour required for "England's continued industrial growth" [Brenner, 1976, p.67] through the $17^{th}$ Century.

## 2 Research Questions

The material above motivates three specific, testable questions, which the model of §3 is built to answer and §3.15 operationalises into concrete outputs. A wider set of consistency checks the model should also satisfy – but which test whether the model behaves sensibly rather than contributing anything new – are collected separately in §2.1.

**RQ1: Is "improvement" a consequence of market exposure, or a precondition of it?** The abstract's central claim is that improvement ideology is as much a *consequence* of newly created market imperatives as it is a cause of dispossession. The model operationalises this as an endogenous tenant "improving disposition" $\iota_j^T(t)$, driven by exposure to competitive rent rather than assumed (§3.8). The falsifiable prediction is that $\iota_j^T(t)$ and reinvestment $k_j(t)$ should rise sharply in the periods *following* a tenant's own conversion to Leasehold, not before it, with tenants who remain Customary throughout showing no comparable rise. If the model instead shows reinvestment rising in anticipation of, or independently of, market exposure, the mechanism would be contradicting the claim it was built to test.

**RQ2: Does conversion spread outward from a small, spatially localised shock, and which channel drives it?** Given uniform inflationary pressure, fiscal deficits accrue to every landlord at roughly the same rate; the theory's claim that the transition begins from a small, localised competitive seed (§3) rather than already existing everywhere requires conversion to spread outward from the ecological seed of §3.2 with a distance-dependent lag, not occur simultaneously. This is the model's central internal-validity test (§3.7): the three transmission channels – tenant mobility, observed neighbour outcomes, and ideological legitimation – are implemented as independent switches precisely so the question "which, if any, is necessary to produce contagion rather than simultaneity" has a direct answer rather than being assumed.

**RQ3: Does the state–landlord alliance matter, as Brenner's England/France comparison claims?** Brenner's comparative argument treats the presence or absence of an independent political ally for the peasantry as the decisive condition separating England's outcome from France's (§3.5). The model's most direct test of this specific claim, as distinct from the internal class-conflict mechanism alone, is whether reversing the state–landlord alliance parameter $\theta$ to the France regime prevents the Leasehold triad from emerging and instead resolves the same fiscal pressure into entrenched Freehold smallholding (§3.6), rather than merely into a slower version of the same England-regime outcome.

### 2.1 Robustness and macro-consistency checks

The four items below are not, on their own, sources of novelty: they ask whether the model behaves sensibly given everything above, in the way a calibration or sanity check would, rather than posing a question the paper is specifically designed to answer. They are collected here, separately from RQ1–RQ3, so that a negative result on one of them is read as a modelling problem to fix, not as a finding about the theory.

- *Sufficiency of the aggregate transition.* Starting from a Customary-dominated initial condition, does the England-regime run move toward Leasehold dominance at the population level at all (§3.15(i))? This is a precondition for RQ1 and RQ2 to be meaningful, not itself a claim under test.
- *Farm-size concentration.* Does the Gini coefficient of tenant holding size rise over the run, and is this attributable to engrossment (§3.9) specifically, rather than to ordinary population turnover (§3.15(ii))?
- *Landless-pool composition, wages and aggregate output.* Does the landless pool show both continued agricultural hiring and a growing cumulative exit to industry, does the wage series diverge from rents in the direction the historical record suggests, and is aggregate output's qualitative shape consistent with the secondary-source series cited in §3.15 (targets (iv) and (vi))?
- *Enclosure as a separable channel.* Is the enclosure share $\Xi(t)$ empirically distinguishable in its timing from rent-conversion-driven dispossession, or does it collapse into the same process (§3.10, §3.15(vii))?

## 3 Model Design

The model is designed as a test of the internal consistency of the Brenner–Wood account, rather than as a stylised illustration of it. The question it asks is whether the class-conflict mechanisms Brenner and Wood identify are jointly sufficient to generate, from a small and spatially localised seed, the macroscopic outcomes the theory is invoked to explain: the collapse of customary tenure, the concentration of landholding, the emergence of the landlord/capitalist-tenant/wage-labourer triad [Brenner, 1976, p.63], and the spread of an "improvement" ethic that legitimises enclosure [Wood, 2002, p.108-109]. Every modelling choice below reflects a specific claim in the primary literature, cited with a page reference, so that the model's assumptions remain traceable to – and falsifiable against – the theory it is testing.

### 3.1 Agents

**Landlord** $i \in \{1, \ldots, N_L\}$. Each landlord holds an estate of tenancies $\mathcal{T}_i$, a wealth stock $W_i(t)$, and a fixed consumption requirement $C_i$ tied to the reproduction of aristocratic class position rather than to profit-maximisation as such [Brenner, 1976, p.31]. Landlords do not choose to become capitalists; they act defensively, to avoid losing their own class position, and any systemic shift toward market dependence is a by-product of that defensive action, both lords and peasants "involuntarily setting in train a capitalist dynamic while acting in class conflict with each other to reproduce themselves as they were" [Wood, 2002, p.52]. Each landlord also has an awareness radius over neighbouring estates' rents and outcomes [Wood, 2002, p.100-101], and an exposure to "improvement" ideology $\iota_i(t)$.

**Tenant** $j$. Rather than instantiating landlord, capitalist-tenant and wage-labourer as three separate agent classes from the outset – which would hard-code the very triad the theory is meant to explain – a tenant is a single agent type carrying a tenure state $\sigma_j(t) \in \{\text{Customary}, \text{Leasehold}, \text{Freehold}, \text{Landless}\}$ that evolves endogenously, and a holding $\mathcal{P}_j(t)$ – a set of land parcels, since holdings themselves grow and shrink through engrossment (§3.9) rather than being fixed at one parcel per tenant. This reflects Brenner and Wood's own emphasis that the triad is an *outcome* of class conflict, not a precondition of it [Brenner, 1976, p.63]; Freehold is included as the historical alternative outcome of the same conflict that prevailed in France rather than England (§3.6).

**Land** parcel $k$, held by a tenant, with a fertility/productivity state $\phi_k(t)$ subject to an exogenous ecological-demographic process (§3.4). Land is not a decision-maker, but its local, heterogeneous condition is what gives the model's initial "seed" of competition a non-arbitrary spatial location, rather than requiring the modeller to plant it by hand.

### 3.2 Spatial structure: an England-shaped lattice

Awareness radius, parcel adjacency, and the "distance-dependent lag" validation test (§3.15) all presuppose an explicit space. Rather than an abstract torus, the lattice is laid over the real outline of England and its heterogeneity is drawn from a real, if necessarily modern, proxy for regional land quality – so that the model's "small, spatially localised seed" (§3) is a location the data itself supplies, not one the modeller plants by hand. England, rather than Great Britain, is the deliberate extent: $\theta$, the scenario parameter that does the real explanatory work in §3.5, stands for a specifically English alignment of Crown and gentry that Brenner and Wood both treat as sui generis to England [Brenner, 1976, Wood, 2002, p.68-75], and Scotland in particular had its own distinct legal and tenurial history through this period; including it would blur the very comparison the model is built to make, of which the France regime (§3.5) is already the intended counterfactual.

**Construction (a one-off preprocessing step, not a runtime dependency).** Land parcels $k$ occupy cells of an $L \times L$ lattice laid over England's bounding box, with cells falling outside the country's boundary [Office for National Statistics, 2022] excluded outright – no wraparound: unlike the earlier torus, a real coastline and the Scottish and Welsh borders are historically meaningful edges, not artifacts to be engineered away. Each surviving cell is assigned, once, to whichever of England's 39 historic counties its centroid falls within, and inherits that county's fertility baseline $\bar{\phi}^{\text{county}}$, computed by area-weighting Natural England's

Agricultural Land Classification (ALC) grades [Natural England, 2021] within the county (Grade 1, "excellent", mapped to the highest score, Grade 5, "very poor", to the lowest). This lookup of ≈40 numbers, one per historic county, is all the running simulation ever needs – the GIS work of intersecting ALC polygons with county boundaries happens once, offline, in producing it. Each parcel then draws

$$\bar{\phi}_k \sim \mathcal{U}\big(\bar{\phi}^{\text{county}(k)} - \zeta,\ \bar{\phi}^{\text{county}(k)} + \zeta\big), \tag{1}$$

retaining fine-grained local heterogeneity within a county (so engrossment and turnover still vary parcel-to-parcel, §3.9) while centring it on a real regional value: modern ALC grades reflect a soil-and-climate endowment that changes on a geological, not a century, timescale, so treating it as invariant over the model's ∼150-year window is a stated proxy assumption, distinct from the fast-moving, endogenous depletion and regrowth (3) already gives each parcel. The model's low-fertility "seed" is then simply wherever this construction happens to put the worst county – historically the upland, wood-pasture and moorland counties rather than the arable South and East – with no separate seed-radius parameter required.

$N_L$ landlord seed points are scattered uniformly at random over the surviving England-shaped cell set (uniform rather than weighted toward more heavily manorialised regions, since weighting would require a second historical dataset – population or manor density circa 1600 – and would start to blur the model from a theory-test with a realistic backdrop into a historical reconstruction, which §3 explicitly disclaims). Estates are then formed by a Voronoi tessellation around these seeds exactly as before: parcel $k$ belongs to landlord $i$'s estate $\mathcal{T}_i(0)$ if $i$'s seed is $k$'s nearest seed among those falling in England. This gives each landlord a contiguous but irregularly shaped estate without requiring the modeller to hand-draw manor boundaries – a representational choice, not a historical claim, in the same sense that the logistic form of (3) is a standard functional choice rather than one drawn from Brenner or Wood.

Two distinct neighbour relations are read off the same lattice, since Brenner and Wood's own language keeps them at different levels: engrossment operates "within the individual tenancy relation" (§3.10), i.e. between adjacent *parcels* under a single landlord, whereas the spread channels of §3.7 operate between *estates*.

- **Parcel adjacency** $\mathcal{N}(k)$ (§3.9): the Moore (8-connected) lattice neighbours of $k$ that belong to the same landlord's estate as $k$.
- **Landlord awareness radius** $\mathcal{N}_i$ (§3.5, §3.7): all landlords $i' \neq i$ whose seed point lies within Euclidean distance $r$ of $i$'s seed point.

The validation test of §3.15(iii) regresses each parcel's first-conversion time on its lattice distance from the centre of the lowest-fertility county identified above.

### 3.3 Causal structure

Figure 1 summarises how the mechanisms below connect; citations for each node are given in the corresponding subsection of the text. Two features are worth flagging before the equations. First, there are three, not one, channels by which a single landlord's break from custom propagates to others – direct tenant mobility, observed neighbour outcomes, and the slower diffusion of a legitimating ideology – implemented as separable switches so that their relative importance can be tested rather than assumed. Second, the diagram makes explicit that *why* a landlord can break custom at all is treated as a structural, largely exogenous condition of the English case (the state–landlord alliance $\theta$), not something the model derives from local bargaining – following Brenner's own comparative argument (§3.5).

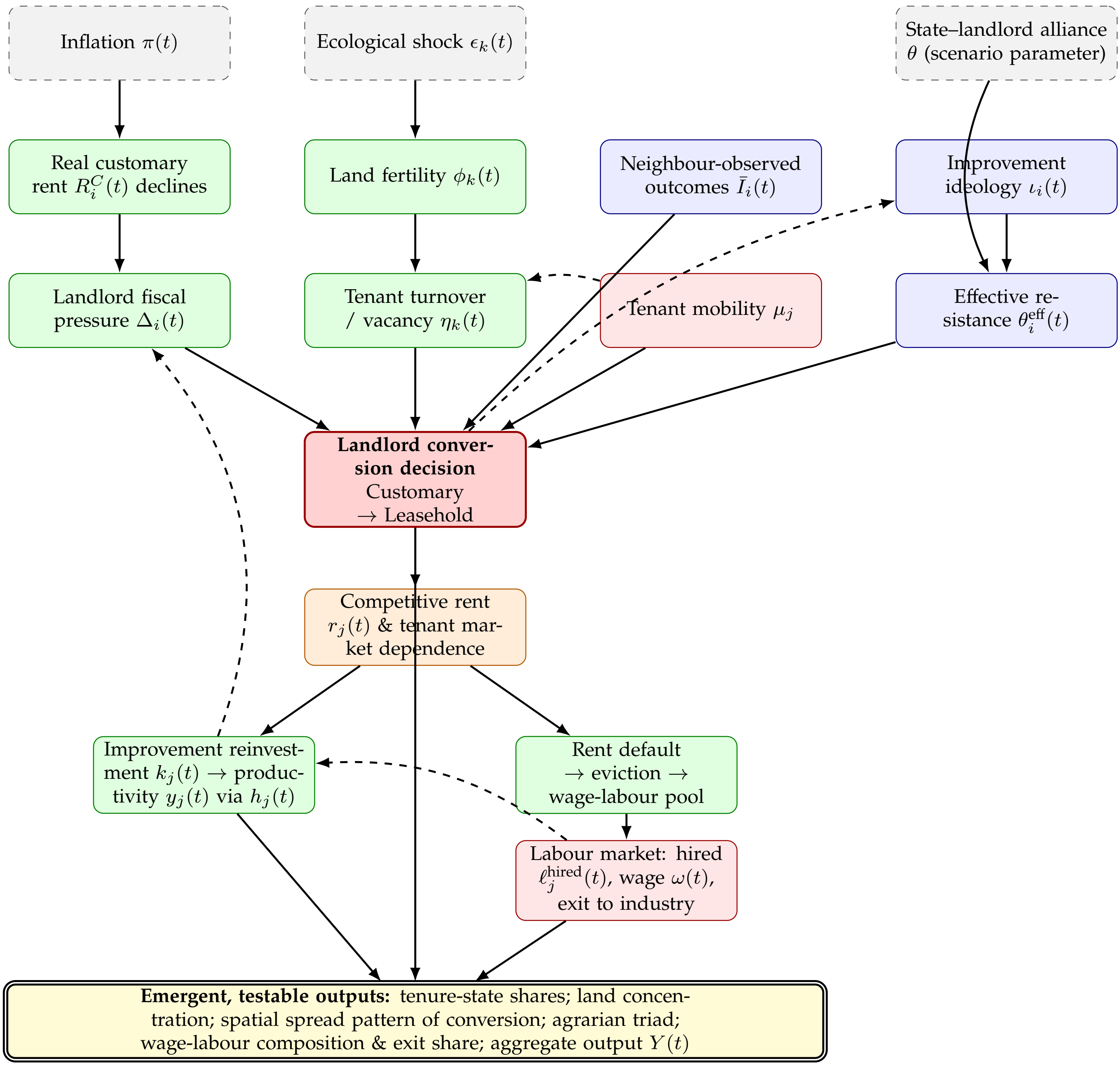


Figure 1: Causal structure of the model. Dashed boxes are exogenous/scenario parameters; solid green boxes are endogenous processes; blue boxes are the three spread channels of §3.7; the red box is the landlord's discrete decision; the double yellow box lists the emergent quantities used as validation targets in §3.15. Dashed arrows denote feedback loops (mobility creating a further vacancy; a conversion seeding the ideology that lowers resistance to the next conversion; tenant improvement reducing the landlord's own fiscal pressure; hired labour from the wage-labour pool raising the hiring tenant's own productivity, §3.12). Freehold entrenchment (§3.6) and enclosure of the commons (§3.10) are omitted from the diagram for legibility and are given only in equations.

### 3.4 Exogenous drivers: why feudalism cannot reproduce itself

Two exogenous processes represent the standing pressure that makes the customary order unsustainable on its own terms, independent of any competitive dynamic.

**Inflation.** Customary rent $r_j^C(0)$ is fixed in nominal terms at the point a tenancy is granted and is not renegotiated while a tenant remains in the Customary state: "rent *per se* (*redditus*) was fixed by custom, and subject to declining long-term value in the face of inflation" [Brenner, 1976, p.61]. With an exogenous price index $\Pi(t) = \prod_{s=1}^{t}(1 + \pi(s))$, the real value of landlord $i$'s customary income is

$$R_i^C(t) = \sum_{j \in \mathcal{T}_i^C(t)} \frac{r_j^C(0)}{\Pi(t)}, \tag{2}$$

which decays monotonically even though nothing in the local land market has changed.

**Ecological-demographic cycle.** Land fertility is modelled as a renewable stock subject to logistic regrowth toward a parcel-specific carrying capacity $\bar{\phi}_k$ – the standard formalism for a resource that grows fastest at intermediate stock levels and saturates as it approaches its ecological ceiling [Verhulst, 1838], here combined with an extraction/shock term exactly as in the bioeconomic model of a harvested renewable resource [Schaefer, 1954] – punctuated by exogenous shocks $\epsilon_k(t)$:

$$\phi_k(t+1) = \phi_k(t) + \rho\, \phi_k(t)\Big(1 - \frac{\phi_k(t)}{\bar{\phi}_k}\Big) - \epsilon_k(t). \tag{3}$$

Neither Brenner nor Wood specifies a functional form for soil-fertility dynamics – nor, on inspection, does Jason W. Moore's account of feudalism's own ecological contradictions [Moore, 2003], which is historical-theoretical rather than mathematical. Their evidence is qualitative, concerning the exhaustion of arable land under insufficient manuring and the extension of cultivation onto marginal soils, and it is this qualitative claim, not a specific equation, that (3) is chosen to instantiate: the lord's surplus extraction "tended to confiscate not merely the peasant's income above subsistence...but at the same time threatened the funds necessary to refurbish...productivity" [Brenner, 1976, p.48], producing a "vicious cycle of the destruction of the peasants' means of support" in which "the crisis of productivity led to demographic crisis" [Brenner, 1976, p.49-50]. The heterogeneity of $\bar{\phi}_k$ across parcels is what gives the model's initial shock a location: some estates are ecologically buffered and some are not, mirroring this uneven pattern of subsistence crisis. A tenancy's turnover hazard rises when output falls below a subsistence threshold $\bar{y}$,

$$\eta_k(t) = \eta_0 + \eta_1 \max\big(0,\ \bar{y} - y_k(t)\big), \tag{4}$$

so that vacancies – through death, flight, or failure to hold the tenancy – cluster where ecological conditions are worst, exactly as Brenner describes for the fourteenth–fifteenth century demographic collapse: "the demographic collapse...left vacant many former customary peasant holdings. It appears often to have been possible for the landlord simply to appropriate these and add them to his demesne" [Brenner, 1976, p.61].

### 3.5 The landlord's conversion decision

At each vacancy, landlord $i$ chooses between renewing the tenancy on customary terms or converting it to a leasehold let at a competitive rent. Define fiscal pressure as the gap between the landlord's fixed consumption requirement and realised real rental income,

$$\Delta_i(t) = C_i - R_i^C(t) - R_i^L(t). \tag{5}$$

The conversion probability at a vacancy is

$$p_i^{\text{convert}}(t) = \sigma\Big(\alpha_0 + \alpha_1 \Delta_i(t) + \alpha_2 \bar{I}_i(t) - \alpha_3\, \theta_i^{\text{eff}}(t)\Big), \tag{6}$$

with $\sigma(\cdot)$ the logistic function, $\bar{I}_i(t)$ the neighbour-observation term (§3.7), and $\theta_i^{\text{eff}}(t)$ the effective strength of institutional resistance a tenant can muster against conversion (§3.7). This is also the point at which landlords historically had a second, non-confrontational lever besides outright conversion: raising the entry fine charged at conveyance, since "there often remained one crucial loophole open to the landlord...He could insist on the right to charge fines...whenever peasant land was conveyed...in the end entry fines appear to have provided the landlords with the lever they needed to dispose of customary peasant tenants, for in the long run fines could be substituted for competitive commercial rents" [Brenner, 1976, p.62]; the model treats a fine escalation as a special case of the same conversion decision, since its economic effect – a rent that tracks the market rather than custom – is identical.

Crucially, $\theta$ is not derived endogenously from a submodel of peasant revolt. Brenner's own comparative argument treats the presence or absence of an independent political ally for the peasantry as the decisive, largely exogenous condition distinguishing England from France: in France "the monarchy...had an interest in limiting the landlords' rents...and thus in intervening against the landlords to end peasant unfreedom and to establish and secure peasant property" [Brenner, 1976, p.69], whereas in England the same centralising process ran through, not against, the landlord class – "important sections of the English nobility and gentry were willing to support the monarchy's centralizing political battle against the disruptive activities of the magnate-warlords...but it was precisely these same landlord elements who were most concerned to undermine peasant property" [Brenner, 1976, p.71-72]. Wood's account of the English state makes the same point institutionally: the aristocracy was "demilitarized before any other aristocracy in Europe...part of the increasingly centralized state...[but] did not possess autonomous 'extra-economic' powers...to the same degree as their continental counterparts" [Wood, 2002, p.99], so that landlords "depended less on their ability to squeeze more rents out of their tenants by direct, coercive means than on their tenants' success in competitive production" [Wood, 2002, p.100]. Accordingly $\theta$ is set as a scenario-level parameter – an "England" regime and a counterfactual "France" regime, used for validation in §3.15 – rather than as an emergent variable, with individual conversion attempts still subject to idiosyncratic stochastic resistance through the logistic error term, representing the real but ultimately unsuccessful local resistance of the sort seen in Ket's Rebellion of 1549: "if successful, the peasant revolts of the sixteenth century...might have 'clipped the wings of rural capitalism'...But they did not succeed" [Brenner, 1976, p.62-63].

### 3.6 Freehold tenure and the closure of an alternative path

The conversion decision above is not the only possible outcome of a customary tenant's struggle against a landlord under fiscal pressure. In the mid-fifteenth century, before landlords found the entry-fine loophole, the outcome was genuinely open: "peasant tenants at this time were striving hard for full and essentially freehold control over their customary tenements, and were not far from achieving it" [Brenner, 1976, p.61]. Freehold is therefore included as a fourth tenure state, not merely as a residual category, because the theory's comparative logic requires a place for the path England did *not* take.

**Initial condition, not emergent outcome.** Consistent with Wood's account, the initial share of Freehold tenants is set low and is not itself a target the England-regime run is expected to explain: "the concentration of English landholding meant that an unusually large proportion of land was worked not by peasant-proprietors but by tenants...This was true even before the waves of dispossession...associated with 'enclosure"' [Wood, 2002, p.99-100], in contrast to France, which "remained a country of peasant proprietors" while "land in England was concentrated in far fewer hands" [Wood, 2002, p.133].

**Freeholders still compete.** Owning land outright does not exempt a tenant from the market imperative, since the compulsion runs through the goods market as much as the rental market: "yeomen were typically the very kind of capitalist tenants who were subject to competitive pressures, and even owner-occupiers would be subject to those pressures once the competitive productivity of agrarian capitalism set the terms of economic survival" [Wood, 2002, p.54]. A Freehold tenant's wealth therefore evolves by the same unified rule as any other tenure state ((13), §3.8, with $\rho_j(t) = 0$), using the same production function (17) and the same competitively-driven improving disposition $\iota_j^T(t)$ and reinvestment rate $\bar{s}\,\iota_j^T(t)$ (§3.8, (15)), with the estate's leasehold benchmark $\hat{r}_i(t)$ standing in as $j$'s shadow rent for the purposes of $M_j(t)$ even though none is actually paid; a Freeholder who cannot sustain $c_j$ for $\tau$ periods sells up and enters the Landless state, while one who prospers can absorb a neighbouring vacant parcel through the same engrossment competition available to capitalist tenants (§3.9) – "the same would increasingly be true even of landowners working their own land. In this competitive environment, productive farmers prospered and their holdings were likely to grow, while less competitive producers went to the wall and joined the propertyless classes" [Wood, 2002, p.103].

**A rare, closing pathway.** At a vacancy where the sitting customary tenant's resistance succeeds (§3.5), there is a further, small probability that the outcome entrenches as Freehold rather than reverting to Customary,

$$p_j^{\text{Freehold}}(t) = \sigma\big(\lambda_0 + \lambda_1\,\theta - \lambda_2\,\mathbb{1}[t \geq t^*]\big), \tag{7}$$

increasing in the same state–landlord alliance parameter $\theta$ used throughout, and falling discretely once the entry-fine loophole fully closes at the same $t^*$ that marks the end of state resistance to enclosure (§3.10) – formalising the closing-off already asserted qualitatively below. In the England regime ($\theta$ low) this is close to zero even before $t^*$; in the France regime ($\theta$ high) this pathway dominates, giving the model's England/France comparison (§3.15) a mechanism rather than only a suppressed probability – the same underlying pressure resolves to the Leasehold triad in one regime and to entrenched Freehold smallholding in the other, exactly the divergence Brenner draws out at length in his own comparison of the two countries' seventeenth-century agrarian structures [Brenner, 1976, p.68-75].

### 3.7 Spread of the market imperative

Given uniform inflationary pressure, fiscal deficits accrue to all landlords roughly together; what produces a spreading, rather than simultaneous, pattern of conversion is that converting is a costly institutional break, so landlords are heterogeneously reluctant to be first. Three channels transmit a successful defection to its neighbours.

*Observed outcomes.* Landlords within $i$'s awareness radius $\mathcal{N}_i$ update on the realised wealth gain of neighbours in proportion to how much of those neighbours' own estates have already converted – a landlord's estate converts tenancy-by-tenancy, not all at once (§3.11), so the signal is a continuous Leasehold share $s_{i'}^L(t) = |\mathcal{T}_{i'}^L(t)|/|\mathcal{T}_{i'}(t)|$ rather than a binary "has converted" flag,

$$\bar{I}_i(t) = \frac{1}{|\mathcal{N}_i|} \sum_{i' \in \mathcal{N}_i} s_{i'}^L(t) \cdot \frac{W_{i'}(t)}{W_{i'}(0)}, \tag{8}$$

with $t = 0$ the simulation's start, at which $W_{i'}(0) > 0$ by construction (§3.14), and $|\mathcal{T}_{i'}(t)|$ in $s_{i'}^L(t)$ counting only currently active, separately-let tenancy slots on estate $i'$ – a parcel absorbed by engrossment (§3.9) has ceased to be its own tenancy and drops out of both numerator and denominator. This directly reflects the historical practice Wood describes of landlords' surveyors explicitly pricing the gap between fixed and competitive rent – "we can watch the development of a new mentality by observing the landlord's surveyor as he computes the rental value of land...against the actual rents being paid by customary tenants...[calculating] 'the annual value beyond rent' or 'value above the oulde [sic] rent"' [Wood, 2002, p.101].

*Tenant mobility.* A Customary tenant $j$ on landlord $i$'s estate migrates to an adjacent, already-converted estate $i' \in \mathcal{N}_i$ with an open Leasehold vacancy (§3.11) with probability

$$p_j^{\text{migrate}}(t) = \mu_j \cdot \bar{I}_i(t) \cdot \mathbb{1}\big[\exists\, i' \in \mathcal{N}_i :\ \text{open Leasehold vacancy at } i'\big], \tag{9}$$

reusing the same observed-outcomes signal $\bar{I}_i(t)$ that drives the landlord's own conversion decision (§3.5): the same surveyor's calculation of "the annual value beyond rent" that recalibrates a landlord's resistance is exactly what a mobile tenant is also responding to. Where more than one $i' \in \mathcal{N}_i$ qualifies, $j$ migrates to $\arg\max_{i'} s_{i'}^L(t) \cdot W_{i'}(t)/W_{i'}(0)$ – the same per-neighbour term summed inside $\bar{I}_i(t)$, i.e. the single most attractive already-converted neighbour rather than an unspecified choice among them. Migration vacates $j$'s parcels at $i$ as a crisis vacancy (§3.11), directly imposing a vacancy on the losing landlord, and admits $j$ to the vacancy at $i'$ under the standard Landless-pool entry terms of §3.11, with capital left behind and reset to $k^{\text{trad}}$ on arrival – the same dispossession of accumulated improvement already described for outright eviction, "the peasant, expropriated and cast adrift" (§1.1), here imposed by the tenant's own choice to leave rather than by the landlord's hand: "success would breed success, and competitive farmers would have increasing access to even more land, while others lost access altogether" [Wood, 2002, p.100-101].

*Ideological legitimation.* Adoption of "improvement" as a legitimating ethic diffuses across landlords as a Bass-type contagion process,

$$\iota_i(t+1) = \iota_i(t) + \beta_1\big(1-\iota_i(t)\big) + \beta_2\,\bar{\iota}_{\mathcal{N}_i}(t)\big(1-\iota_i(t)\big), \tag{10}$$

and lowers the effective force of institutional resistance faced at conversion,

$$\theta_i^{\text{eff}}(t) = \theta\cdot\big(1-\xi_0-\xi_1\iota_i(t)\big). \tag{11}$$

This is the mechanism Wood documents in detail: enclosure and the extinguishing of customary right were pursued not only economically but legally and rhetorically, since "between the sixteenth and eighteenth centuries, there was growing pressure to extinguish customary rights that interfered with capitalist accumulation...traditional conceptions of property had to be replaced by new, capitalist conceptions of property" [Wood, 2002, p.108], with "reasons of 'improvement'...cited systematically as the basis of title to property and as grounds for extinguishing traditional rights" [Wood, 2002, p.115], of which Locke's labour theory of property was the paradigm articulation [Wood, 2002, p.108-112].

These three channels are implemented as independent switches so that the paper's Results section can ask which, if any, is necessary to reproduce a spatially spreading – rather than simultaneous – pattern of conversion: the model's central internal-validity test.

### 3.8 Tenant response: market dependence and improvement

Every occupied tenancy, whatever its tenure state, faces a state-dependent rent

$$\rho_j(t) = \begin{cases} r_j^C(0)/\Pi(t) & \sigma_j(t) = \text{Customary}, \\ \hat{r}_i(t) & \sigma_j(t) = \text{Leasehold}, \\ 0 & \sigma_j(t) = \text{Freehold}, \end{cases} \tag{12}$$

and wealth evolves by a single rule regardless of state,

$$w_j(t+1) = w_j(t) + y_j(t) - \rho_j(t) - c_j, \tag{13}$$

with $c_j$ a fixed subsistence requirement; (13) nests both the Leasehold and Freehold cases of earlier drafts as $\rho_j(t) = \hat{r}_i(t)$ and $\rho_j(t) = 0$ respectively. The Leasehold rent is not individually negotiated with the sitting tenant but set once for the whole estate at landlord $i$'s local competitive benchmark – a single going rate a tenant has no part in setting, which is what makes it genuinely competitive rather than a personalised squeeze on whoever happens to be struggling,

$$\hat{y}_i(t) = \frac{1}{|\mathcal{T}_i^L(t)|}\sum_{j\in\mathcal{T}_i^L(t)} y_j(t), \qquad \hat{r}_i(t) = \theta_{\text{rent}}\,\hat{y}_i(t-1), \qquad \theta_{\text{rent}} \in (0,1), \tag{14}$$

(bootstrapped, at a landlord's first-ever conversion, from the estate's own customary rent scale in place of an as-yet-undefined $\hat{y}_i(t-1)$), so that every Leasehold tenant on estate $i$ faces the same $\rho_j(t) = \hat{r}_i(t)$, aggregating to $R_i^L(t) = |\mathcal{T}_i^L(t)|\cdot\hat{r}_i(t)$ used in $\Delta_i(t)$ (§3.5); no inflation discount is needed here since $\hat{r}_i(t)$ already re-prices with current output, unlike $r_j^C(0)$, which is fixed in nominal terms at grant. This is exactly the calculation Wood attributes to the landlord's own surveyor (§3.7): $\hat{r}_i(t)$ is the "rental value of land...against the actual rents being paid" [Wood, 2002, p.101], made explicit as a number rather than left as a comparison the landlord alone draws.

**Improvement as a response to competitive necessity, not a precondition of it.** Rather than gating reinvestment on tenure state by fiat, the model makes a tenant's willingness to reinvest an endogenous "improving disposition" $\iota_j^T(t) \in [0,1]$ – the superscript distinguishing it from the landlord's legitimating ideology $\iota_i(t)$ of §3.7, a conceptually distinct variable that happens to answer the same abstract-level question (why does anyone start improving?) for the other class of agent – driven by exactly the pressure the theory identifies as its cause: the gap between what a tenant's land could competitively fetch and what they actually produce,

$$M_j(t) = \begin{cases} \rho_j(t)/y_j(t) & \sigma_j(t) \in \{\text{Customary}, \text{Leasehold}\}, \\ \hat{r}_i(t)/y_j(t) & \sigma_j(t) = \text{Freehold}, \end{cases} \qquad \iota_j^T(t+1) = \iota_j^T(t) + \beta^T M_j(t)\left(1 - \iota_j^T(t)\right), \quad (15)$$

with reinvestment scaling continuously in the disposition it has produced rather than switching on or off at conversion,

$$k_j(t+1) = k_j(t) + \bar{s}\,\iota_j^T(t)\max\left(0,\, w_j(t+1) - w^{\min}\right), \quad (16)$$

and every tenant initialised at $\iota_j^T(0) = 0$ and a common traditional capital baseline $k_j(0) = k^{\text{trad}}$, so Customary and newly-converted tenants start from the same technical position and diverge only as $M_j(t)$ diverges. Under Customary tenure $M_j(t)$ is small and falling – the same inflationary erosion of $\rho_j(t)$ already driving landlord fiscal pressure (§3.4) also means a customary tenant feels little competitive necessity to improve, with no separate rule required to switch reinvestment off; at conversion $\rho_j(t)$ jumps from the tenant's own historically low customary rent to the estate's competitive benchmark $\hat{r}_i(t)$, and $M_j(t)$ jumps with it, so the "new mentality" documented by Wood's surveyors takes hold at exactly the moment the tenant becomes market-dependent, not before. This directly operationalises the same market compulsion already used to justify Freeholders' continued exposure (§3.6): "even owner-occupiers would be subject to those pressures once the competitive productivity of agrarian capitalism set the terms of economic survival" [Wood, 2002, p.54].

Because $\hat{r}_i(t)$ tracks the estate's own competitive peers rather than any individual tenant's realised output, the reverse of (15) also holds: a tenant who does not raise $\iota_j^T(t)$ – and hence $k_j(t)$ and $y_j(t)$ – falls behind a rent they had no part in setting, and sustained shortfall triggers the same $\tau$-period eviction rule as rent default generally (below), vacating the parcel for engrossment by a more successful neighbour (§3.9). Reinvestment is therefore not only individually adaptive under pressure but also the criterion by which the tenant population is selected – compete, in the sense of (15), or be outcompeted, in the sense of §3.11.

This raises output through a diminishing-returns function of the capital and labour applied across the tenant's whole holding $\mathcal{P}_j(t)$, not a single fixed parcel,

$$y_j(t) = \Phi_j(t)^{\mu} \cdot k_j(t)^{\gamma} \cdot h_j(t)^{1-\mu-\gamma}, \qquad \mu, \gamma \in (0,1),\ \mu + \gamma < 1, \quad (17)$$

where $\Phi_j(t) = \sum_{k \in \mathcal{P}_j(t)} \phi_k(t)$ is the fertility-weighted size of the holding and $h_j(t) = \bar{\ell} + \ell_j^{\text{hired}}(t)$ is total labour applied: the tenant's own household endowment $\bar{\ell}$ plus any hired labour $\ell_j^{\text{hired}}(t)$ drawn from the landless pool (§3.12). How $\mathcal{P}_j(t)$ itself grows is the subject of §3.9; here it is enough that this is the mechanism, not merely an analogy, that Brenner identifies as distinguishing English agrarian development: the displacement of "the traditionally antagonistic relationship in which landlord 'squeezing' undermined tenant initiative, by an emergent landlord-tenant symbiosis which brought mutual co-operation in investment and improvement" [Brenner, 1976, p.65]. A tenant of any tenure state who cannot cover $\rho_j(t) + c_j$ for $\tau$ consecutive periods loses the tenancy and enters the Landless state – not to leave the model, but to become the labour supply the surviving tenants in (17) draw on, as §3.12 sets out; this is a crisis-type vacancy in the sense of §3.11, with no heir to fall back on, and is the mechanism by which a tenant who fails to develop $\iota_j^T(t)$ (above) is removed from the model.

### 3.9 Land concentration: the engrossment of holdings

Neither Brenner nor Wood treats landlord-level estate size as something the sixteenth–seventeenth century dynamics generate: Wood is explicit that big-landlord concentration was already in place before the story starts – "land in England had for a long time been unusually concentrated, with big landlords holding an unusually large proportion, in conditions that enabled them to use their property in new ways" [Wood, 2002, p.99]. Landlord estate size $|\mathcal{T}_i(0)|$ is therefore initialised as a heterogeneous but fixed starting condition (§3.1), not an emergent output. What the theory does explain endogenously is a different margin entirely: the concentration of *farms*, i.e. tenant holdings, as customary smallholdings are merged into fewer, larger units –

> "with the peasants' failure to establish essentially freehold control over the land, the landlords were able to *engross, consolidate and enclose*, to create large farms and lease them to capitalist tenants who could afford to make capitalist investments" [Brenner, 1976, p.63],

a pattern Brenner also documents in the comparative section via Bowden: "under the stimulus of growing population, rising agricultural prices, and mounting land values...large estates were built up at the expense of small holdings" [Brenner, 1976, p.42].

At any vacated parcel $k$ – whether from the ecological-turnover channel of §3.4 or a rent-default eviction (§3.12) – the landlord's decision is therefore not only *whether* to convert (§3.5) but *to whom* to let it: a new, separate tenant, or an existing neighbouring tenant's holding. Writing $\mathcal{N}(k)$ for the Leasehold or Freehold tenants already adjacent to $k$ and $j^* = \arg\max_{j\in\mathcal{N}(k)} k_j(t)$ for the most capital-intensive of them, the probability of engrossment into $j^*$ rather than separate re-letting is

$$q_i(t) = \sigma\big(\psi_0 + \psi_1\Delta_i(t) + \psi_2\, k_{j^*}(t)\big), \tag{18}$$

reusing the landlord's fiscal pressure $\Delta_i(t)$ (§3.5) and the neighbour's own capital stock $k_j(t)$ (§3.8) rather than introducing new state. If engrossment occurs, $\mathcal{P}_{j^*}(t+1) = \mathcal{P}_{j^*}(t) \cup \{k\}$, directly operationalising Wood's point that "success would breed success, and competitive farmers would have increasing access to even more land, while others lost access altogether" [Wood, 2002, p.100-101]. If not, the parcel enters the ordinary conversion decision of §3.5 as before.

Because the same displaced smallholder who loses the parcel typically becomes the labour the engrossing tenant now needs to work the larger holding (§3.12), the model produces tenant-farm concentration and the wage-labour pool as two faces of a single event, rather than as separately assumed processes.

### 3.10 Enclosure of the commons

Engrossment operates parcel by parcel, within the individual tenancy relation. Enclosure is a distinct, slower process that operates on shared subsistence resources external to any single tenancy: "there existed common lands, on which members of the community might have grazing rights or the right to collect firewood, and there were various other kinds of use rights on private land, such as the right to collect the leavings of the harvest during specified periods of the year" [Wood, 2002, p.107]. Enclosure is the extinguishing of exactly these rights, not merely the fencing of land: "enclosure meant not simply a physical fencing of land but the extinction of common and customary use rights on which many people depended for their livelihood" [Wood, 2002, p.108], and its early, most disruptive wave targeted pasture for sheep – Thomas More, "though himself an encloser, described the practice as 'sheep devouring men"' [Wood, 2002, p.108-109].

Let $\Xi(t) \in [0,1]$ denote the aggregate share of common/use-right land already enclosed, diffusing alongside improvement ideology – Wood ties the two processes together directly, since it is "reasons of 'improvement"' that were "cited systematically...as grounds for extinguishing traditional rights" [Wood, 2002, p.115] – with a discrete acceleration once state resistance to enclosure disappears:

$$\Xi(t+1) = \Xi(t) + \nu_1\big(1-\Xi(t)\big) + \nu_2\,\bar{\iota}(t)\big(1-\Xi(t)\big) + \nu_3\mathbb{1}[t \geq t^*]\big(1-\Xi(t)\big), \tag{19}$$

where $\bar{\iota}(t)$ is the population-average of the ideology term $\iota_i(t)$ (§3.7) and $t^*$ is a scenario parameter marking the point at which the state stops contesting enclosure at all: "in its earlier phases, the practice was to some degree resisted by the monarchical state...but once the landed classes had succeeded in shaping the state to their own changing requirements – a success more or less finally consolidated in 1688...there was no further state interference" [Wood, 2002, p.109], the same alignment of state and landlord interest already used to set $\theta$ (§3.5) – Brenner independently ties enclosure to the identical alliance, describing the English gentry as the very group "most concerned to undermine peasant property in the interests of enclosure and consolidation" [Brenner, 1976, p.71-72].

Enclosure's effect is external to the tenancy whose rent status changes: it extends the turnover hazard (4) with a third term, raising it for *any* nearby smallholder – Customary tenant or marginal Freeholder alike – who depended on commons access to make ends meet, whether or not their own parcel is the one being converted.

A fixed share $\rho_{\text{commons}}$ of parcels in each estate is flagged commons-dependent at $t = 0$, chosen at random independently of $\bar{\phi}_k$, and carries $\mathbb{1}[k \text{ commons-dependent}] = 1$ for the life of the run – operationalising Wood's "various other kinds of use rights" (quoted above) as a fixed structural feature of the manor, not one that itself grows or is assigned endogenously,

$$\eta_k(t) = \eta_0 + \eta_1 \max\big(0, \bar{y} - y_k(t)\big) + \eta_2\, \Xi(t) \cdot \mathbb{1}[k \text{ commons-dependent}], \tag{20}$$

which nests (4) exactly when $\Xi(t) = 0$. This is what lets the model represent enclosure as the specific driver of dispossession *without* an individual tenancy conversion – the historical "growing plague of vagabonds, those dispossessed 'masterless men' who wandered the countryside" [Wood, 2002, p.108-109] – as distinct from, though reinforcing, the rent-conversion and engrossment channels above.

### 3.11 Vacancy resolution and population continuity

The mechanisms above are all triggered "at a vacancy" without yet specifying, for every vacancy, in what order a landlord's options are exercised, nor where the next occupant of a re-let tenancy comes from. Both questions are answered by distinguishing two kinds of vacancy already implicit in (4): the baseline hazard $\eta_0$, representing ordinary generational succession (the death of a tenant with an heir), and the excess hazard – $\eta_1 \max(0, \bar{y} - y_k(t))$, $\eta_2\, \Xi(t) \cdot \mathbb{1}[\cdot]$, rent-default eviction under any tenure state (§3.8), or a tenant's own departure by migration (§3.7) – representing genuine dispossession with no customary claimant left to fall back on. This is exactly the distinction Brenner draws between ordinary conveyance, at which the landlord's only historical lever was the entry fine [Brenner, 1976, p.62], and demographic-collapse vacancies that a landlord "could simply appropriate...and add...to his demesne" [Brenner, 1976, p.61] because no one remained to contest them.

**Succession vacancies** (probability $\eta_0$ at any occupied parcel, each period). The tenancy is not structurally vacant: an heir succeeds the outgoing tenant by default, inheriting $\mathcal{P}_j(t)$ and a fraction $\xi_{\text{inherit}}$ of their wealth, with capital reset to the traditional baseline $k^{\text{trad}}$ (§3.8). This is precisely the moment of "conveyance" at which the landlord may still act (§3.5): the engrossment check and conversion decision below run exactly as at a crisis vacancy, so an entry fine can still be levied on the heir even though the parcel never stood empty.

**Crisis vacancies.** No customary claimant remains, so parcel $k$ genuinely falls to the landlord's disposal. Resolution proceeds in a fixed order:

1. *Engrossment check* (§3.9): draw engrossment with probability $q_i(t)$. If it succeeds, $k$ is absorbed into $j^*$'s holding and resolution ends here – no new tenant is created.

2. *Conversion decision* (§3.5): if not engrossed, draw conversion with probability $p_i^{\text{convert}}(t)$.

3. *Freehold diversion* (§3.6): if conversion is resisted (the draw in step 2 fails), draw $p_j^{\text{Freehold}}(t)$; on success the new occupant enters directly as Freehold rather than Customary.

4. *New occupant.* If neither engrossed nor converted, a new Customary tenant takes up $k$ at $t+1$ at the parcel's unchanged nominal rent $r_j^C(0)$ – custom attaches to the land, not the person. If converted (step 2) or diverted to Freehold (step 3), the new occupant is drawn preferentially from the Landless pool $\mathcal{L}(t)$ (§3.12), subject to an entry cost $\chi \cdot \hat{r}_i(t)$ – the same entry-fine object introduced qualitatively in §3.5, now given a magnitude, where $\hat{r}_i(t)$ is estate $i$'s own competitive rent benchmark (§3.8), the same figure the new tenancy will itself be charged if Leasehold. A Landless agent takes the vacancy if $w_\ell(t) \geq \chi\, \hat{r}_i(t)$, entering with $w_j(0) = w_\ell(t) - \chi\, \hat{r}_i(t)$, $k_j(0) = k^{\text{trad}}$, and $\iota_j^T(0) = 0$.

If step 4 requires a Landless entrant and none clears the entry cost, $k$ joins a vacant-parcel queue, re-attempting steps 1–4 (with $\Delta_i(t)$ and $q_i(t)$ recomputed) every subsequent period until filled; a parcel in this queue produces no output and no rent, itself contributing to $\Delta_i(t)$. Land held by neither a tenant nor the queue does not otherwise exist in the model: every parcel is, at all times, either held, being engrossed, or in the queue.

This closes the model's population accounting without an unmodelled reservoir of prospective tenants: the customary sector reproduces itself indefinitely through succession, matching the historical persistence of the customary peasantry across the period Brenner and Wood describe, while the Leasehold/Freehold sector's growth is bounded by the size of the Landless pool it itself generates through eviction and migration (§3.7, §3.8) – the "reserve army" channel of §3.12 is therefore also, endogenously, the channel that staffs new capitalist tenancies, not merely the wage-labour market.

### 3.12 The wage-labour pool: hiring, wages, and exit

A landless agent is not a sink. Brenner's consolidation mechanism only works because the larger holdings it produces need more hands to farm than a single household supplies: demesne land was "leased out to larger tenants...[who] in turn sub-let to smaller farmers or employed wage labourers" [Brenner, 1976, p.61]. The model must therefore specify what the landless do next, not merely how tenants come to lose their land.

**Labour demand.** A Leasehold or Freehold tenant's demand for hired labour grows with the capital they have committed to the parcel, once it exceeds what their own household can work,

$$\ell_j^{\text{demand}}(t) = \max\big(0,\ \delta\, k_j(t) - \bar{\ell}\big), \tag{21}$$

so that it is precisely the successful, improving, consolidating tenants – Wood's "success would breed success" [Wood, 2002, p.100-101] – who hire, and unsuccessful ones who are hired. Demand is a claim on a finite pool, not a guarantee: if aggregate demand $D(t) = \sum_j \ell_j^{\text{demand}}(t)$ exceeds the landless pool $|\mathcal{L}(t)|$, hiring is rationed pro rata,

$$\ell_j^{\text{hired}}(t) = \ell_j^{\text{demand}}(t) \cdot \min\Big(1,\ \frac{|\mathcal{L}(t)|}{D(t)}\Big), \tag{22}$$

so that $h_j(t) = \bar{\ell} + \ell_j^{\text{hired}}(t)$ in (17) never draws on more bodies than actually exist, and the unmet portion of demand – rather than a phantom labour supply – is exactly what the wage adjustment below prices.

**Wage determination.** The same excess demand $D(t) - |\mathcal{L}(t)|$ that rationing absorbs in quantity is what moves price: aggregate demand $D(t)$ is matched against the size of the landless pool $|\mathcal{L}(t)|$ through a simple tâtonnement,

$$\omega(t+1) = \omega(t)\Big(1 + \kappa\,\frac{D(t) - |\mathcal{L}(t)|}{|\mathcal{L}(t)|}\Big), \tag{23}$$

so that a landless pool growing faster than tenant demand for hired labour depresses the wage – the model's "reserve army" channel, consistent with the historical pattern of rents rising while wages stayed comparatively flat over the same period [Kerridge, 1953, Clark, 2002].

**Landless income and exit.** A landless agent $\ell$ sells its labour endowment at the going wage,

$$w_\ell(t+1) = w_\ell(t) + \omega(t)\,\bar{\ell} - c_\ell, \tag{24}$$

and if $w_\ell(t) < 0$ for $\tau'$ consecutive periods, exits the rural model entirely rather than persisting as an idle agent. This is not a modelling convenience but the mechanism Brenner names directly: "agricultural improvement not only made it possible for an ever greater proportion of the population to leave the land to enter industry" [Brenner, 1976, p.67]. In the reverse direction, a landless agent whose wealth clears the entry cost $\chi \cdot \hat{r}_i(t)$ of an open vacancy (§3.11) may re-enter directly as a Leasehold tenant – or, at a succession vacancy left without an heir, as a Customary one – giving the model a (theoretically vanishing, as land concentrates) channel of upward mobility rather than assuming the triad, once formed, is a closed caste system.

The point of specifying this rather than leaving "wage labourer" as a terminal label is that the theory's own account of *why* the transition matters beyond agriculture runs through exactly this population: rising agricultural labour-productivity is what "created a highly productive agriculture capable of sustaining a large population not engaged in agricultural production, but also an increasing propertyless mass that would constitute both a large wage-labour force and a domestic market for cheap consumer goods – a type of market with no historical precedent. This is the background to the formation of English industrial capitalism" [Wood, 2002, p.103]. The exit channel above is what lets the model, in principle, speak to that claim rather than only to the agrarian triad in isolation.

### 3.13 Model schedule

The mechanisms above are specified as update rules for individual state variables; what has not yet been fixed is the order in which they run within a single period $t \to t+1$, nor the bookkeeping state – beyond tenure, holdings, wealth and capital – needed to evaluate them. Table 1 lists every piece of state actually required, including the consecutive-shortfall counters $d_j(t)$ and $d_\ell(t)$ that the $\tau$- and $\tau'$-period eviction and exit rules (§3.8, §3.12) presuppose but do not themselves name. The list below then fixes the order. All quantities dated $t$ (rents, fiscal pressure, the observed-outcomes and ideology terms, wage) are computed once at the start of the period from state as of $t$ and held fixed while every within-period event is resolved – a standard synchronous-update convention, adopted so that two vacancies arising on the same estate in the same period are resolved against the same $\Delta_i(t)$ rather than one seeing a value the other has already changed.

| Agent | State | Meaning |
|---|---|---|
| Landlord $i$ | $W_i(t)$, $\iota_i(t)$ | wealth; legitimating ideology exposure (§3.7) |
| | $\mathcal{T}_i(t)$ | set of currently active, separately-let tenancy slots on the estate |
| Tenant $j$ | $\sigma_j(t)$, $\mathcal{P}_j(t)$ | tenure state; set of held parcels |
| | $w_j(t)$, $k_j(t)$, $\iota_j^T(t)$ | wealth; capital; improving disposition (§3.8) |
| | $d_j(t)$ | consecutive periods $w_j(t)$ has been short of $\rho_j(t)+c_j$; eviction at $d_j(t)=\tau$ |
| Landless $\ell$ | $w_\ell(t)$, $d_\ell(t)$ | wealth; consecutive periods with $w_\ell(t) < 0$; exit at $d_\ell(t)=\tau'$ |
| Parcel $k$ | $\phi_k(t)$, $\bar{\phi}_k$ | current fertility stock; fixed carrying capacity (§3.4) |
| | commons-flag | fixed at $t=0$ (§3.10) |

Table 1: State carried by each agent type, in addition to the fixed per-agent parameters of Table 2.

1. **Exogenous update.** Draw $\pi(t)$ and update $\Pi(t)$; draw $\epsilon_k(t)$ for every parcel. Since $\phi_k(t+1)$ ((3)) depends only on $\phi_k(t)$ and $\epsilon_k(t)$, it may be computed at any point in this list.
2. **Labour demand and clearing.** For every Leasehold or Freehold tenant, compute $\ell_j^{\text{demand}}(t)$ from $k_j(t)$; ration to $\ell_j^{\text{hired}}(t)$ against $|\mathcal{L}(t)|$; update $\omega(t+1)$ (§3.12).
3. **Production.** Compute $y_j(t)$ ((17)) for every occupied tenancy from $\Phi_j(t)$, $k_j(t)$ and $h_j(t) = \bar{\ell} + \ell_j^{\text{hired}}(t)$.
4. **Wealth, capital and disposition.** With $\rho_j(t)$ already fixed at the start of $t$ (Customary and Leasehold rents were set at $t-1$; Freehold is always 0), update $w_j(t+1)$ ((13)), $M_j(t)$ and $\iota_j^T(t+1)$ ((15)), and $k_j(t+1)$; update $d_j(t+1)$ (reset to 0 if solvent, incremented otherwise). Landless agents update $w_\ell(t+1)$ and $d_\ell(t+1)$ symmetrically (§3.12).
5. **Landlord fiscal pressure.** Compute $R_i^C(t)$, $R_i^L(t) = |\mathcal{T}_i^L(t)|\,\hat{r}_i(t)$, and $\Delta_i(t)$ (§3.5).
6. **Spread terms.** Compute $\bar{I}_i(t)$ from $s_{i'}^L(t)$ and $W_{i'}(t)$ as of the start of $t$ (§3.7); update landlord ideology $\iota_i(t+1)$ and effective resistance $\theta_i^{\text{eff}}(t)$; update enclosure share $\Xi(t+1)$ (§3.10).
7. **Vacancy determination.** For every occupied parcel, draw a succession event (probability $\eta_0$) or a crisis event (the excess-hazard terms of (20), or $d_j(t+1)=\tau$, or a successful migration draw $p_j^{\text{migrate}}(t)$, §3.7); collect the resulting list of vacancies (§3.11).
8. **Vacancy resolution.** Resolve every vacancy collected in step 7, in random order, by the fixed procedure of §3.11, using the period-$t$ values of $\Delta_i(t)$, $q_i(t)$, $p_i^{\text{convert}}(t)$ and $p_j^{\text{Freehold}}(t)$ computed above – *not* recomputed between vacancies within the same period.
9. **Rent reset.** Recompute $\hat{y}_i(t)$ (§3.8) from this period's realised Leasehold output (step 3) over $\mathcal{T}_i^L(t)$ as it stood during production, giving $\hat{r}_i(t+1) = \theta_{\text{rent}}\hat{y}_i(t)$ for the next period.
10. Advance to $t+1$ with all updated state now current.

### 3.14 Parameters and initial conditions

Table 2 collects every free quantity introduced above with a placeholder starting value. Because §3.15 treats the model's outputs as a qualitative plausibility check rather than a fitted calibration target, these are deliberately round starting points for the first runs and the sensitivity sweeps that follow, not estimates drawn from data.

| Symbol | Value | Meaning |
|---|---|---|
| *Space and population (§3.2)* | | |
| $L$ | 60 | lattice side length ($L \times L$ grid over England's bounding box, non-England cells excluded) |
| $N_L$ | 40 | number of landlords/estates |
| $r$ | 8 | landlord awareness radius |
| $\zeta$ | 2 | within-county fertility jitter (§3.2) |
| *Initial agent state (§3.1)* | | |
| $W_i(0)$ | 100 | landlord initial wealth |
| $C_i$ | $\mathcal{U}(80, 120)$ | landlord consumption requirement |
| initial $\sigma_j$ shares | 0.95 / 0.05 / 0 / 0 | Customary / Freehold / Leasehold / Landless |
| $w_j(0)$ | 10 | tenant initial wealth |
| $k^{\text{trad}}$ | 1 | traditional (non-market) capital baseline |
| $r_j^C(0)$ | $\mathcal{U}(0.5, 1.5)$ | customary rent at grant |
| $\bar{\ell}$ | 1 | household labour endowment |
| $\mu_j$ | $\mathcal{U}(0, 0.3)$ | tenant mobility propensity |
| *Exogenous processes (§3.4)* | | |
| $\pi(t)$ | 0.02 | inflation rate per period |
| $\rho$ | 0.1 | fertility regrowth rate |
| $\bar{\phi}_k$ | county baseline $\pm\zeta$ (§3.2) | carrying capacity, from ALC grades [Natural England, 2021] |
| $\epsilon_k(t)$ | Bernoulli(0.05) $\times\, \mathcal{U}(0, 2)$ | ecological shock |
| $\bar{y}$ | 2 | subsistence output threshold |
| $\eta_0, \eta_1, \eta_2$ | 0.02, 0.05, 0.03 | turnover hazard coefficients |
| *Conversion, resistance and Freehold (§3.5, §3.6)* | | |
| $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ | $-3$, 0.05, 1, 1 | conversion logit coefficients |
| $\theta$ | 2 (England) / 6 (France) | state–landlord alliance regime |
| $\xi_0, \xi_1$ | 0.3, 0.5 | ideology discount on resistance |
| $\lambda_0, \lambda_1, \lambda_2$ | $-4$, 0.3, 1 | Freehold-diversion logit coefficients |
| $t^*$ | 100 | period at which state resistance ends |
| $\chi$ | 2 | entry cost, in periods of rent |
| $\xi_{\text{inherit}}$ | 0.5 | wealth fraction retained at succession |
| *Spread (§3.7)* | | |
| $\beta_1, \beta_2$ | 0.01, 0.05 | ideology diffusion coefficients |
| $\iota_i(0)$ | 0 | initial ideology exposure |
| *Tenant economy (§3.8)* | | |
| $\theta_{\text{rent}}$ | 0.3 | competitive-benchmark extraction share, $\hat{r}_i(t) = \theta_{\text{rent}}\hat{y}_i(t-1)$ |
| $\beta^T$ | 0.1 | necessity-driven growth rate of tenant improving disposition $\iota_j^T(t)$ |
| $\iota_j^T(0)$ | 0 | initial tenant improving disposition (all tenure states) |
| $\bar{s}$ | 0.3 | reinvestment-rate ceiling, scaled by $\iota_j^T(t)$ |
| $w^{\min}$ | 0 | wealth floor below which no reinvestment occurs |
| $c_j$ | 1 | tenant subsistence requirement |
| $\mu, \gamma$ | 0.3, 0.3 | Cobb–Douglas fertility/capital exponents |
| $\tau$ | 3 | periods of default tolerated before eviction |
| *Engrossment and enclosure (§3.9, §3.10)* | | |
| $\psi_0, \psi_1, \psi_2$ | $-2$, 0.05, 0.1 | engrossment logit coefficients |
| $\rho_{\text{commons}}$ | 0.2 | share of parcels flagged commons-dependent |
| $\nu_1, \nu_2, \nu_3$ | 0.01, 0.05, 0.1 | enclosure diffusion coefficients |
| $\Xi(0)$ | 0 | initial enclosure share |
| *Labour market (§3.12)* | | |
| $\delta$ | 0.5 | labour-demand coefficient |
| $\kappa$ | 0.1 | wage adjustment speed |
| $\omega(0)$ | 1 | initial wage |
| $c_\ell$ | 0.8 | landless subsistence requirement |
| $\tau'$ | 3 | periods of negative wealth tolerated before exit |

Table 2: Placeholder parameter values for the first qualitative runs (§3.15); all are candidates for the sensitivity analysis, not fitted estimates.

### 3.15 Emergent outputs and validation targets

Because the model is a validity test rather than an illustration, its outputs are defined as things that could fail to match the theory's claims. At minimum: (i) the share of tenancies in each tenure state over time, now including Freehold alongside Customary, Leasehold and Landless; (ii) *farm*-size concentration – the Gini coefficient of tenant holding size $A_j(t) = |\mathcal{P}_j(t)|$, which the engrossment mechanism of §3.9 predicts should rise over the run – kept *explicitly distinct* from landlord-estate-size concentration, which §3.9 treats as a fixed initial condition rather than an emergent target, so the model is not credited with reproducing a claim the theory itself does not make; (iii) whether conversion spreads outward from the ecological seed with a distance-dependent lag, or occurs simultaneously – the direct test of the "localised seed" claim; (iv) aggregate output $Y(t) = \sum_j y_j(t)$, compared *qualitatively* against the historical rent, wage and grain-output series assembled from secondary sources [Clark, 1991, Clark, 2002, Kerridge, 1953], used as a plausibility check on the shape of the transition rather than as a calibration target; (v) a direct comparative test of Brenner's own argument, by re-running the model with $\theta$ set to a "France" regime (institutional resistance systematically prevails), checking not only whether the Leasehold triad fails to emerge but whether the same displaced pressure instead resolves into entrenched Freehold smallholding (§3.6), as the theory predicts: "in England, it was precisely the absence of such rights that facilitated the onset of real economic development" [Brenner, 1976, p.74-75]; (vi) the composition of the landless pool over time – the share still seeking agricultural hire, the wage series $\omega(t)$ against the historical rent/wage ratio, and the cumulative share that has exited the rural model – as the model's operationalisation of the claim that agrarian improvement is what freed population "to leave the land to enter industry" [Brenner, 1976, p.67]; and (vii) the enclosure share $\Xi(t)$ against the timing of dispossession implied by tenure-state shares, testing whether the model reproduces enclosure and rent-conversion as related but empirically separable drivers of landlessness, rather than a single conflated process.

# 4 Results

# 5 Discussion

# 6 Conclusions

# 7 Acknowledgements

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